MULTI-SCHOOL SECONDARY SCHOOL MANAGEMENT SYSTEM

Project Plan • Core Modules • Chapa Payment Integration • Deployment

Platform	Multi-school secondary school management system
Recommended stack	React + Tailwind CSS, Node.js + Express, PostgreSQL
Payment gateway	Chapa hosted checkout with server-side payment verification
Currency	Ethiopian Birr (ETB)
Merchant ID supplied	388628 (must be confirmed with Chapa for the intended merchant setup)

1. Project Goals

Build a secure platform for multiple secondary schools. Each school should manage its own students, teachers, classes, academic records, invoices, and financial reports, while the Super Admin oversees the platform. School data must be isolated so one school cannot access another school's private records.

2. Main Modules (Steps 1–6)

Step	Module	Scope
1	Secure login & permissions	Authentication, password hashing, sessions/tokens, role-based access, password reset, audit logs.
2	Super Admin dashboard	Register/activate schools, assign school admins, manage platform settings, view high-level statistics.
3	School operations	Students, teachers, classes, subjects, attendance, examinations, grades, report cards.
4	Chapa integration	School-specific payment settings, hosted checkout, transaction references, server-side verification, webhook handling.
5	Finance & receipts	Invoices, partial payments, balances, payment history, downloadable receipts, daily/monthly/yearly reports.
6	Deployment	Production database, environment secrets, HTTPS, backups, monitoring, domain, deployment and acceptance testing.

3. User Roles

Role	Typical permissions
Super Admin	Manage schools, platform configuration, and system-wide oversight.
School Admin	Manage one school's users, classes, academic setup, and school records.
Finance Officer	Create invoices, reconcile verified payments, issue receipts, and view finance reports.
Teacher	Manage assigned classes, attendance, assessments, and marks.
Student	View their own timetable, attendance, grades, invoices, and payment history.
Parent/Guardian	View linked student's progress, invoices, balances, and receipts; initiate payments.

4. Chapa Payment Workflow

1. A school finance officer creates an invoice for a student.

2. The parent/student selects an eligible invoice and chooses “Pay with Chapa.”

3. The backend calculates the payable amount from trusted database records and initializes hosted checkout.

4. The customer completes payment on Chapa's hosted checkout page.

5. The backend verifies the payment with Chapa and checks status, amount, currency, reference, and school/order association.

6. Only after successful verification does the system record the payment, update the invoice balance, and make a receipt available.

Payment security requirements

• Never put Chapa secret keys in frontend code, mobile apps, public repositories, or chat messages.

• Store credentials securely on the server and restrict access to authorized school administrators.

• Do not mark an invoice paid based only on the browser return URL; verify server-side.

• Make webhook handling idempotent and record transaction references to prevent duplicate payment processing.

• Confirm with Chapa that each school can use its own merchant account/API credentials under the intended integration. A Merchant ID alone is not enough to process payments.

5. Recommended Database Entities

Schools, Users, Roles, SchoolMemberships, Students, Parents/Guardians, Teachers, Classes, Subjects, Enrollments, Attendance, Assessments, Grades, Invoices, InvoiceItems, PaymentAttempts, Payments, Receipts, AuditLogs, and SchoolPaymentSettings. Most school-owned records should include a school_id field and be protected by authorization checks.

6. Deployment Checklist

• Configure production environment variables and rotate any exposed secrets.

• Use managed PostgreSQL or a secured production database with automated backups.

• Enable HTTPS and configure allowed frontend origins (CORS).

• Add rate limiting, input validation, logging, monitoring, and secure error handling.

• Test role boundaries and verify that users cannot access another school's records.

• Test successful, failed, cancelled, duplicate, and delayed payment notifications.

• Run end-to-end acceptance tests before launching to real schools.

7. Useful Official Resource

Chapa Developer Documentation: https://docs.chapa.global/docs/v2/getting-started

Project status: This PDF is a planning document. It does not represent a completed, production-ready application. The source code must be implemented, tested, and reviewed before deployment.